

Inner Aspect in Cantonese

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Abstract

The paper provides the first systematic study on the structural properties of inner aspect in Cantonese. In total 14 postverbal inner aspectual elements have been identified (e.g., completive *jyun* in *sik-jyun* ‘finish eating’). It is shown that they occupy a middle position: above resultative verbal complements (e.g., *baau* ‘full’ in *sik-baau* ‘eat and become full’), but below outer aspectual elements (e.g., perfective *-zo* in *sik-zo* ‘ate’). Syntactically, they are lower than *vP* but may be either higher or lower than *VP*. Furthermore, while some of them are phase complements, some other have already become or on their way to be verbal suffixes.

Keywords: inner aspect, phase complement, verbal suffixes, Cantonese

1. Introduction

Aspect refers to “different ways of viewing the internal temporal constituency of a situation” (Comrie 1976:3) denoted by the predicate. Aspect may be divided into situation aspect and viewpoint aspect (Smith 1991, 1997; see Smith 1994 and Soh 2014 for Chinese). The former is usually known as eventualities (‘Aktionsart’), including states, activities, accomplishments, and achievements; and the latter includes the familiar perfective and imperfective aspects.

Structurally speaking, there are at least two aspectual projections on the clausal spine: above *vP*, the outer aspect; and below *vP*, the inner aspect (i.e., “the split aspect approach”, MacDonald 2008, Travis 2010; for Mandarin and/or Cantonese, see Gu 1995, Tsai 2008, Huang, Li, and Li 2009, Sybesma 2017, Lu, Lipták, and Sybesma 2019, Yip 2020a, Tang 2022, Lee and Pan 2024). While attempts have been made to map outer aspect to viewpoint aspect and inner aspect to situation aspect (e.g., Lu, Lipták, and Sybesma 2019), these two classifications do not exactly align. It is possible for viewpoint aspects to be encoded in both inner and outer aspect positions (see the recent study in Yan & Yuan 2024). This study maintains that the situation-viewpoint split is semantic and the inner-outer split is structural/syntactic, and there is no one-to-one mapping between them. The focus of this paper will be the inner-outer split.

Cantonese is rich in postverbal elements pertaining to aspect. There has been a fruitful of work on outer aspectual elements such as perfective suffix *-zo* (咗), experiential suffix *-gwo* (過), progressive suffix *-gan* (緊), durative suffix *-zyu* (住), among others (Lin 1963, Cheung 1972/2007:§4, Gao 1980: 34–59, Matthews and Yip 1994:210–221, Zhu 1994, Li et al. 1995:418–424, 552–565, Chan 1996, Chang 1996, Cheng 2001, Chor 2004; see the comprehensive review in Tang 2015:§7).¹ There is, however, very little attention on the **inner aspect** in Cantonese. Canonical inner aspectual elements include *hou* (好) in *so-hou* (鎖好) ‘locked properly’ or *zoek* (著) in *fan-zoek* (瞓著) ‘slept’, which express the phase/stage of an action and have been called phase complement (PCs) (時相補語) (Chao 1968: 446–450). Some of them have been documented in some of work cited above, but there is, to the best of my knowledge, no systematic study on them.

The primary goal of this paper is to provide a comprehensive description of the structural properties of the inner aspectual elements in Cantonese. As will be discussed in

¹ Cantonese examples are transcribed in *Jyutping* (the Linguistic Society of Hong Kong Cantonese Romanization Scheme, 1993), and tones (1–6) are represented when necessary.

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Section 2, they are grammatically different from resultative verbal complements (RVCs), such as *baau* (飽) ‘full’ in *sikbaau* (食飽) ‘eat and become full’, and from outer aspectual elements such as perfective *-zo* (咗). Morphologically, RVCs are the closest to the verbal root, then inner aspect, and further, outer aspect, and the furthest postverbal element is the universal quantificational suffix *-saai* (晒) (Zhan 1958, T. H. T. Lee 1994, Tang 2003, P. Lee 2004, 2012, *i.a.*), as schematized in (1).

(1) The verbal template in Cantonese

VERBAL ROOT < RVCs < inner aspect < outer aspect < *-saai*

I will proceed to discuss the grammatical status of inner aspectual elements with formal diagnostic tests in Section 3: whether they are verbal complements (i.e., PCs) or verbal suffixes. Section 4 sketches a formal analysis. I propose that outer aspects are *vP*-external and inner aspects are *vP*-internal, and further suggest the formal distinction between verbal complements and suffixes is whether they are base-generated VP-internally or not. Section 5 concludes.

2. Defining inner aspect

Let us become with an operational definition of (postverbal) inner aspectual elements in Cantonese, given in (2). (2)b and (2)d set them apart from RVCs; and (2)c sets them apart from outer aspectual suffixes.

(2) The operational definition of (postverbal) inner aspectual elements

- a. A morpheme following the verbal root; which ($\rightarrow$ postverbal)
- b. Contributes to aspectual meaning; ($\rightarrow$ aspectual)
- c. Can be followed by the perfective suffix *-zo*; and ($\neq$ outer aspect)
- d. Cannot undergo verb doubling alone and/or together with the verbal root. ($\neq$ RVCs)

The definition in (2) gives 14 inner aspectual elements, listed in (3) with the aspectual contribution briefly described. I will go through each part of the definition below.

(3) Postverbal inner aspectual elements in Cantonese

(Cheung 1972/2007:113–115, Matthews and Yip 1994:210–221, Tang 2015:71–89)

- a. *dou2* (到/倒) ‘arrive’: Achievement
E.g., *wan-dou* (搵到) ‘found’ (Fang 2003, Huang 2021, Lai & Pang 2023)
- b. *zoek6* (著) ‘be on target’: Achievement
E.g., *fan-zoek* (瞓著) ‘slept’
- c. *gin3* (見) ‘see (vs. look)’: Achievement, successful perception
E.g., *teng-gin* (聽見) ‘hear (vs. listen)’
- d. *seng4* (成) ‘succeed’: Achievement, successful action
E.g. *joek-seng* (約成) ‘succeeded in making an appointment’
- e. *can1* (親) ‘(lit.) intimate’: Achievement, with adversative results
E.g., *da-can* ‘injure by hitting’ (Gu and Yip 2004, Sio 2020)
- f. *jyun4* (完) ‘finish’: Completive
E.g., *sik-jyun* (食完) ‘finished eating’
- g. *hou2* (好) ‘good’: Completive with a “properly-finished” flavor
E.g., *zou-hou* (做好) ‘being done’
- h. *dim6* (掂) ‘all right’: Completive with a “properly-finished” flavor
E.g., *gaau-dim* (搞掂) ‘finished, settled’

- i. *hei2* (起) ‘finish, ready, (lit.) lift’: Completive with a “ready” flavor
E.g., *waak-hei* (畫起) ‘finished drawing’ (Chor 2007)
- j. *lok6* (落) ‘finish (a long time ago), (lit.) fall’: Completive, temporally distant
E.g., *zyu-lok* (教落) ‘taught way back then’ (Cheng 1998)
- k. *zyu6* (住) ‘hold’: Continuative states, for verbs of attachment
E.g. *kam-zyu* (𨳊住) ‘cover still’
- l. *sat6* (實) ‘firm’: Continuative states
E.g. *mong-sat* (望實) ‘keep looking’
- m. *ding6* (定) ‘(lit.) steady’: Doing something in advance
E.g., *zyu-ding* (煮定) ‘cooked in advance’ (Wong 2018)
- n. *gwo3* (過) ‘(lit.) pass’: Doing something again to fix undesirable outcomes
E.g., *cungsan se-gwo* (重新寫過) ‘write again’ (Chen & Lin 2006, Liu & Yip 2023)

First, all these elements are postverbal (=2a). They encode aspectual meaning (=2b) by either indicating the eventualities or imposing a viewpoint on it, including achievement (a-e), completives (f-j), and continuation of states (k-l).² Some come with additional flavors (mostly from the literal meaning of the morpheme), such as the “properly done” flavor of *hou* (好) ‘(lit.) good’.

There are two special cases: *ding* (定) in (m) with an “in advance” meaning, and *gwo* (過) in (n) with an “again” meaning (to be distinguished with experiential *-gwo* 過). Both of them presuppose some situation/event other than the current one: a prospective event for *ding* (定), for example, cooking in advance of going to work in (m); and a previous event for *gwo* (過), which is also a writing event in (n). Yet, they do not seem to impose a particular viewpoint on the event structure. Nevertheless, they both have requirements on the eventualities of the predicates they attach to: [+durative] and [+dynamic] for *ding* (定) (Wong 2018), and [+dynamic] for *gwo* (過). Thus, I regard them as aspectual elements as well.³

Note that a number of elements in (3) have other uses or homophonous counterparts. For example, *dou* (到) has a capacity modal use, as in *co-dou ng-go jan* (坐到五個人) ‘able to have people sit’ (e.g., Huang 2021). *Zoek* (著) has a suffixal use with unintentionality (Lai and Chin 2018), as in *m-siusam jyu-zoek keoi* (唔小心遇著佢) ‘met him/her unintentionally’. Also, there is another *-can* (親) expressing universal quantification on events like *juk-can zau tung* (郁親就痛) ‘feel hurt once (you) move’ (Zhan 1958, P. Lee 2017, Yip 2020b, 2022). *-Zyu* (住) is usually regarded as a durative aspect suffix as in *sik-zyu faan* (食住飯) ‘eating rice’, but it is a

² How completives can be distinguished from achievement as well as perfective aspect is a non-trivial issue. As well-known, perfective aspect in Chinese only indicates endpoints rather than completion of an event as in (i), unlike English. Similarly, achievement markers like *dou* (到) in (ii) also does not indicate completion. A thorough discussion is however beyond the scope of this paper, and readers are referred to Smith (1994).

(i) 我寫咗一封信，但未寫完。

Ngo se-**zo** jat-fung seon, daan mei se-**jyun**.
1SG write-PFV one-CL letter but not.yet write-COMPL.

Lit.: ‘#I wrote a letter, but haven’t finished writing (it).’ (infelicitous in English)

(ii) 我聽到一首歌，但未聽完(全首)。

Ngo teng-**dou** jat-sau go, daan mei teng-**jyun** cyun-sau.
1SG hear-ACHV one-CL song but not.yet hear-COMPL whole-CL.

‘I heard a song, but haven’t finished hearing/listening (the whole song).’

³ In traditional grammar (e.g., Cheung 1972/2007, Gao 1980, Chang 1994), “again” is often regarded as a type of aspect, termed repetitive aspect (重行體/多回體), and relatedly resumptive aspect (回復體), such as *-faan* (翻). For a comparison between *-faan* and repetitive *-gwo*, see Liu & Yip (2023).

phase complement when with “verbs of attachment”, like *kam* (𨮒) ‘cover’ and *pou* (抱) ‘hug’ (a distinction dated back to Cheung 1972/2007:158-162).⁴ *Hei* (起) has a special converbal use requiring a following clause as in *gong-hei zau nau* 講起就癲 ‘become mad once mentioning it’ (see Tang 2018, 2022; Yip 2019, 2022 for converbal suffixes in Cantonese), where *hei* indicates inchoative aspect. Similarly, *lok6* conveys inchoative aspect when used in middle-like constructions like *sik-lok m-co* 食落唔錯 ‘it tastes (lit. eats) good’ (Cheng 1998, Shi 2003, N. Lee 2016). On the other hand, *ding* (定) has a RVC use as in *kei-ding(-ding)* (企定/企定定) ‘stand steadily’ (Wong 2018). Lastly, it is well known that *-gwo* (過) is also an experiential suffix as in *lai-gwo* 嚟過 ‘have come before’, an outer aspectual element. This paper only focuses on the inner aspectual use and remains neutral to whether these other uses are polysemous (i.e., still the same morpheme) or homophonous (i.e., already a different morpheme).

The definitions (2)a-b (postverbal elements with aspectual contribution) are necessary conditions rather than sufficient ones for inner aspectual elements. While the primary meaning RVCs is the lexical meaning, such as *baau* (飽) ‘full’ in *sikbaau* (食飽) ‘eat and become full’ and *sei* ‘die’ (死) in *dasei* (打死) ‘beat to death’, they have a (secondary) aspectual restriction [+telic] (either accomplishment or achievement). The restriction, however, stems from the very nature of resultatives: having a result entails a natural endpoint of the event. The restriction is also related to the eventualities of the RVCs themselves. For example, since *sei* (死) is an instant event (or a change of state), *dasei* (打死) is achievement, as opposed to accomplishment *sikbaau* (食飽) where *baau* (飽) is a stative predicate. In contrast, the inner aspectual elements in (3) do not lexically encode a (clear) result of the event (even for *-can*, whose result is underspecified), but they primarily indicate the temporal structure of a situation. There is hence a principled difference between the aspectual contribution by inner aspect and by RVCs.

Second, regarding definition (2)c, all inner aspectual elements can be followed by the perfective suffix *-zo* (咗), just like RVCs, such as *sik-baau-zo* (食飽咗) ‘eaten and became full’ (see Cheung 1972/2007). What (2)c differentiates is the position of inner aspect from that of outer aspect: they occupy different postverbal slots, which structurally correspond to different syntactic positions. The examples are given in (4).⁵

(4) *-Zo* (咗) suffixation

- a. 阿明攞到咗一個 offer。

(Huang 2021:4)

Aaming lou-**dou-zo** jat-go ofa.

Ming get-ACHV-PFV one-CL offer

‘Ming got an offer.’

⁴ There is a similar split for Mandarin *-zhe*, see Yuan (1993) and Tsai (2008).

⁵ The source URL links of the internet examples are provided below, accessed on 2022-6-23.

(c):<https://www.youtube.com/watch?v=57s8tFuEXbU>

(d):<https://www.facebook.com/stationerylo/photos/a.852095568464119/1314818108858527/?type=3>

(g):<https://pages.facebook.com/1835535343424298/photos/a.1856762237968275/2779070852404071/?type=3&source=48>

(h):<https://words.hk/zidin/%E6%8E%82>

(i):https://m.facebook.com/szetosifu/photos/a.276853775694116/1097213856991433/?type=3&locale2=zh_CN

(j):https://maylibibi.blogspot.com/2009/07/bibi_2573.html

(k):<https://www.am730.com.hk/%E5%A8%B%E6%A8%82/%E6%BB%95%E9%BA%97%E5%90%8D%E8%B7%8C%E6%96%B75%E6%A2%9D%E8%82%8B%E9%AA%A8-%E9%9D%A0%E9%A3%9F%E5%97%8E%E5%95%A1%E6%AD%A2%E7%97%9B%E9%96%8B%E5%B7%A5-/39339>

(l):https://m.facebook.com/szetosifu/photos/a.276853775694116/1097213856991433/?type=3&locale2=zh_CN

(m):<https://es.restaurantguru.com/%E6%99%AE%E6%B4%B1-Hong-Kong>

- b. 佢瞓著咗。 (Cheung 2007:114)
Keoi fan-**zoek-zo**.
3SG sleep-ACHV-PFV
'He has slept.'
- c. 佢係夢見咗自己係一隻蝴蝶嗎? (Video caption, 2021-2-10)
Keoi hai mung-**gin-zo** zigei hai jat-ze wudip maa3?
3SG be dream-ACHV-PFV self be one-CL butterfly SFP
'Did he dream that he became a butterfly?'
- d. 又係一個傾下傾下又傾成咗嘅 Project 啦。 (Social media, 2020-10-8)
Jauhai jat-go king-haa-king-haa jau king-**seng-zo** ge pozek laa1.
again.be one-CL talk-PROG-talk-PROG again talk-ACHV-PFV GE project SFP
'Just another (business) project that we managed to strike over time.'
- e. 你嚇親咗佢。 (Matthews and Yip 1994:228)
Nei haak-**can-zo** keoi.
2SG scare-ACHV-PFV 3SG
'You scared her.'
- f. 食完咗飯好耐嘞! (Cheung 2007:149)
sik-**jyun-zo** faan hou-noi lo3!
eat-compl-PFV rice very-long SFP
'(I) have finished my meal for quite some time!'
- g. 以為做好咗心理準備。 (SocialMedia, 2021-6-23)
Jiwai zou-**hou-zo** samlei zeonbei.
think do-COMPL-PFV mental preparation
'(I) thought (I was) mentally prepared.'
- h. 單 case 我搞掂咗㗎嘞。 (Words.hk)
Daan case ngo gaau-**dim-zo** gaa3 laak3.
CL case 1SG tackle-COMPL-PFV SFP SFP
'I have finished this case.'
- i. 我幅畫終於畫起咗。 (SocialMedia, 2016-3-17)
Ngo fuk waak zungjyu waa-**hei-zo**
1SG CL drawing eventually draw-COMPL-PFV
'My painting is finally done.'
- j. 我返工之前 (...) 煮落咗湯底。 (Blog, 2009-7-12)
Ngo faangung zicin (...) zyu-**lok-zo** tongdai.
1SG go.to.work before cook-COMPL-PFV soup.base
'Before going to work, I have already cooked the broth.'
- k. 肋骨插住咗個肺。 (News, 2020-2-6)
Laakgwat caap-**zyu-zo** go fai.
rib puncture-CONT-PFV CL lung
'a rib punctured the lung and got stucked.'
- l. 難以置信咁望實咗自己部手機。 (Creative writing, 2021-12-26)
Naanjiziseon-gam mong-**sat-zo** zigei bou saugei.
unbelievable-ly stare-CONT-PFV self CL cell.phone
'(He) stares at his cell phone unbelievably.'
- m. 同埋好夠熱，唔會好似煮定咗好耐。 (Food guide, 2022-4-27)
Tungmaai hou gau-jit, m-wu houci zyu-**ding-zo** hou-noi.
and very enough-hot not-will seem.like cook-in.advance-PFV very-long
'It's also hot enough and doesn't look like being prepared a long time ago.'

- n. 重新寫過咗《拜仁頌》嘅第三段歌詞。 (Liu & Yip 2023:101)
Cungsan se-**gwo-zo** Baaijanzung ge daisaam dyun goci.
again write-again-PFV Bayernhymne GE third line lyrics
'Rewrote the third line of the lyrics of *Bayernhymne*.'

Outer aspectual suffixes, on the other hand, cannot be stacked (Cheung 1972/2007). Progressive *-gan* (緊) cannot be suffixed by *-zo* (咗) (for obvious reasons of aspectual conflicts), as well as experiential *-gwo* (過), as shown in (5)a-b.⁶ Recall that *-zyu* (住) has an outer aspectual counterpart—this durative *-zyu* (住) also resists *-zo* (咗) suffixation, as in (5)c.

(5) Failure of *-zo* (咗) suffixation

- a. *佢已經睇緊咗呢本書。
Keoi jiging tai-**gan-zo** ni-bun syu.
3SG already read-PROG-PFV this-CL book
Int.: 'S/he has been already reading this book.'
- b. *佢已經睇過咗呢本書。
Keoi jiging tai-**gwo-zo** ni-bun syu.
3SG already read-EXP-PFV this-CL book
Int.: 'S/he has been already read this book.'
- c. *佢已經食住咗飯。
Keoi jiging sik-**zyu-zo** faan.
3SG already read-DUR-PFV rice
Int.: 'S/he has been already eating.'

Last but not least, the definition (2)d, verb doubling, serves as an extremely powerful tool to tease inner (and outer) aspectual elements apart from RVCs. To begin with, verbs in Cantonese may be doubled for contrastive topicality (T. T.-M. Lee 2022, 2024a).

(6) 做，佢係會做嘅。

Zou, keoi hai wui **zou** ge2.
do 3SG be will do SFP
'As for doing, s/he will do.'

(T. T.-M. Lee 2024a:43)

The doubled verbs must be “bare” and cannot carry the outer aspectual suffix *-zo* along in (7)a-b (T. T.-M. Lee 2024b). In contrast, RVCs can be carried along, as in (7)b.

(7) Contrast in verb doubling: outer aspect vs. RVCs

- a. {食/*食咗}，佢係已經食咗嗰碗飯嘅。
{**Sik/ *sik-zo**}, keoi hai jiging **sik-zo** go-wun faan ge2.
eat eat-PFV 3SG be already eat-PFV that-CL rice SFP
'As for eating, s/he has already eaten that bowl of rice.'

⁶ This is different from Mandarin, where experiential *-guo* (過) can be followed by (suffixal) perfective *-le* (了):

- (i) 他已經看過了那本書了
Ta yijing kan-**guo-le** nei-ben shu le
3SG already read-EXP-PFV that-CL book SFP
'S/he has already read that book.'

(Gu 1995:52)

- b. {食/食飽/*食飽咗}, 佢係已經食飽咗嘅。
{**Sik/ sik-baau/ *sik-baau-zo**}, keoi hai jiging **sik-baau-zo** ge2.
eat eat-full eat-full-PFV 3SG be already eat-full-PFV SFP
‘As for eating (and becoming full), s/he has already eaten and become full.’

Importantly, inner aspectual elements are degraded to be doubled along. This is illustrated with *dou* (到) (for achievement), *jyun* (完) (for completives), *zyu* 住 (for continuatives), *ding* (定), and *gwo3* (過) in (8). Some of them are arguably more acceptable than doubling with suffixes, but the crucial point is that they are still not as natural as doubling with RVCs in (7)b.

(8) Failure of doubling along with the verb

- a. {執/??執到}, 佢尋日係執到一張銀紙嘅。
{**Zap/ ??zaap-dou**}, keoi camjat hai **zaap-dou** jat-zoeng nganzi ge2.
pick pick-ACHV 3SG yesterday be pick-ACHV one-CL bill SFP
‘As for picking up, s/he has picked up a money bill yesterday.’
- b. {食/*食完}, 佢係已經食完個碗飯嘅。
{**Sik/ *sik-jyun**}, keoi hai jiging **sik-jyun** go-wun faan ge2.
eat eat-COMPL 3SG be already eat-COMPL that-CL rice SFP
‘As for eating, s/he has finished eaten that bowl of rice.’
- c. {抱/?抱住}, 個媽媽係抱住個 B 嘅。
{**Pou/ ?pou-zyu**}, go mama hai **pou-zyu** go bi ge2.
hold hold-CONT that mother be hold-CONT CL baby SFP
‘As for holding, the mother is holding the baby.’
- d. {行/??行定}, 佢係有行定入去先嘅。
{**Haang/??haang-ding**}, keoi hai jau **haang-ding** japheoi sin1 ge2.
walk walk-in.advance 3SG be have walk-in.advance inside SFP SFP
‘As for walking, s/he has walked inside (there) in advance.’
- e. {寫/*寫過}, 佢係已經重新寫過篇文嘅。
{**Se/ *se-gwo**}, keoi hai jiging cungsan **se-gwo** pin man ge2.
write write-again 3SG be already again write-again CL paper SFP
‘As for writing, s/he has already written the article again.’

An even more striking contrast is whether the postverbal elements may be doubled alone without the verbal roots.⁷ While it is possible for RVCs as in (9)a, it is clearly impossible for inner aspectual elements in (9)b, letting alone outer aspectual suffixes in (9)c.

(9) Contrasts in doubling alone without the verb

- a. 飽, 佢係已經食飽咗嘅。
Baau, keoi hai jiging **sik-baau-zo** ge2.
full 3SG be already eat-full-PFV SFP
‘As for becoming full, s/he has already eaten and become full.’
- b. *完, 佢係已經食完個碗飯嘅。
***Jyun**, keoi hai jiging **sik-jyun** go-wun faan ge2.
COMPL 3SG be already eat-COMPL that-CL rice SFP
‘As for eating, s/he has finished eaten that bowl of rice.’

⁷ I thank Anonymized for drawing for attention to this contrast in Mandarin.

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- c. *咗，佢係已經食咗嗰碗飯嘅。
*-**Zo**, keoi hai jiging **sik-zo** go-wun faan ge2.
PFV 3SG be already eat-PFV that-CL rice SFP
‘As for eating, s/he has already eaten that bowl of rice.’

Note that the failure to double alone is a general property of all inner aspectual elements regardless of their grammatical status. As will be discussed in Section 3, some of them (e.g., *zoek* 著) are verbal complements rather than suffixes. As shown in (10), however, all the inner aspectual elements cannot be doubled alone without the verb.

(10) Failure to double alone without the verb

- a. {夢/*到/*見}，佢係尋晚夢{到/見}老豆嘅。
{**Mung/*dou/*gin**}, keoi hai cammaan **mung{-dou/gin}** loudau ge2.
dream ACHV 3SG be last.night dream-ACHV dad SFP
‘As for dreaming, s/he dreamt of Dad last night.’
- b. {瞓/*著}，佢係瞓著咗嘅。
{**Fan/*zoek**}, keoi hai **fan-zoek-zo** ge2.
sleep ACHV 3SG be sleep-ACHV-PFV SFP
‘As for sleeping, s/he has fallen asleep.’
- c. {傾/*成}，佢係傾成咗個 project 嘅。
{**King/*seng**}, keoi hai **king-seng-zo** go pozek ge2.
talk ACHV 3SG be talk-ACHV-PFV CL project SFP
‘As for bargaining, s/he has successfully bargained for the project.’
- d. {嚇/*親}，我係嚇親咗佢嘅。
{**Haak/*can**}, ngo hai **haak-can-zo** keoi ge2.
scare ACHV 1SG be scare-ACHV-PFV 3SG SFP
‘As for scaring, I have scared him/her.’
- e. {做/*好/*掂/*起}，佢係已經做{好/掂/起}呢樣嘢。
{**Zou/*hou/*dim/*-hei**}, keoi hai jiging **zou{-hou/dim/hei}** ni-joeng laak3
do COMPL 3SG be already do-COMPL this-CL SFP
‘As for doing, s/he has already done this.’
- f. {教/*落}，老豆係一早有教落嘅，佢係……
{**Gaau/*lok**}, loudau hai jatzou jau **gaau-lok** ge2, daanhai...
teach COMPL dad be early have teach-COMPL SFP but
‘As for teaching, Dad did teach about this long time ago, but...’
- g. {望/*住/*實}，佢係有望{住/實}你嘅，佢係……
{**Mong/*zyu/*sat**}, keoi hai jau **mong{-zyu/sat}** nei ge2, daanhai...
watch CONT CONT 3SG be have watch-CONT 2SG SFP but
‘As for staring, s/he is staring at you, but ...’
- h. {行/*定}，佢係有行定入去先嘅。
{**Haang/*-ding**}, keoi hai jau **haang-ding** japheoi sin1 ge2.
walk in.advance 3SG be have walk-in.advance inside SFP SFP
‘As for walking, s/he has walked inside (there) in advance.’
- i. {寫/*過}，佢係已經重新寫過篇文嘅。
{**Se/*-gwo**}, keoi hai jiging cungsan **se-gwo** pin man ge2.
write again 3SG be already again write-again CL paper SFP
‘As for writing, s/he has already written the article again.’

Table 1 summarizes the contrasting properties of inner aspectual elements with RVCs and outer aspectual elements.

Postverbal elements	Attached to a verbal root?	Primarily aspectual meaning?	- <i>Zo</i> suffixation?	Doubling (alone and/or with the verb)?
RVCs	YES	NO (only secondary)	YES	YES
Inner aspect	YES	YES	YES	NO
Outer aspect	YES	YES	NO	NO

Table 1: Contrasting properties of inner aspect with RVCs and outer aspect

3. Phase complement or verbal suffixes?

After identifying the inner aspectual elements, the next question is what their grammatical status is. As elements attaching to the verbal root, there are at least two possibilities: verbal complements or verbal suffixes. Usually, most inner aspectual elements are regarded as the former as phase complements, PCs. Nevertheless, as we will see below, some should be treated (and even have been treated in the literature) as verbal suffixes.

The traditional criteria for verbal suffixes are listed in (11), from Cheung (1972/2007) and have been influential in Cantonese linguistics (e.g., adopted by P. Lee 2004, 2012 and Tang 2015). Note that some of the criteria might be questionable, such as (11)d. Suffix stacking is indeed possible in Cantonese with *-saai* (晒) (Tang 2003, 2015), such as *sik-gwo-saai* (食過晒) ‘has eaten all’. I thus discard this criterion.⁸

(11) The traditional criteria for verbal suffixes from Cheung (1972/2007)

- Productivity/flexibility: suffixes can combine with a wide range of verbal roots.
- Boundness: suffixes are bound morphemes.
- Vacuity: suffixes lack lexical meaning.
- Exclusivity: suffixes cannot be stacked. *(not considered in this paper)*
- Potential form: suffixes cannot participate in V-*dak/m*-X forms.
- Following VO and VR compounds: suffixes can attach to Verb-Object and Verb-Resultative compounds. (i.e., suffixes may follow RVCs.)

Let us begin with (11)a. Some inner aspectual elements like *dou* (到), *jyun* (完), *ding* (定) and *gwo* (過) can attach to almost any [+dynamic] and/or [+durative] predicates (i.e., excluding stative *hung* 紅 ‘be red’ and instant *sei* 死 ‘die’). *Hou* (完) and *dim* (掂) are also quite productive, but cannot attach to involitional or uncontrollable predicates: compare *fan-jyun* (瞓完) ‘finished sleeping’ and **fan-hou/dim* (*瞓好/掂). *Hei* (起) and *lok* (落) only attach to transitive verbs, but not all of them; **ji-hei/lok* (*醫起/落) ‘(Int.) healed someone’ is not acceptable.

Other elements are however very restricted. For example, *gin* (見) only attaches to verbs of perception such as *tai-gin* (睇見) ‘saw’, *mong-gin* (望見) ‘saw’, *teng-gin* (聽見) ‘heard’, and even reject some other perceptual verbs like **man-gin* (*聞見) ‘(Int.) smelled’. *Zoek* (著) similarly is very limited and almost only attaches to verbs of sleeping. *Zyu* (住) and *sat* (實)

⁸ While (11)f holds almost across-the-board, there also a potential counter-example to (11)e: *-saai* can participate in potential forms, such as *sik-dak/m-saai* (食得/唔晒) ‘can/cannot eat (them) all’. Nevertheless, *-saai* is arguably a verbal suffix (see extensive arguments in P. Lee 2012:26-30).

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only combine with a few verbs of “attachment” as discussed above. Some are not as unproductive as these four, but still quite limited. *Seng* (成) is also an achievement marker like *dou* (到), but it cannot attach some of the verbs *dou* (到) can attach to, like **zap-seng* *執成 ‘(Int.) successfully pick something up’. *Can* (親) usually attaches to verbs with a specific manner (Gu & Yang 2004). The inner aspectual elements are roughly classified into four groups according to their productivity in (12).

(12) Productivity of inner aspectual elements

- a. Very productive (4): *dou* (到), *jyun* (完), *ding* (定), *gwo* (過)
- b. Relatively productive (4): *hou* (好), *dim* (掂), *hei* (起), *lok* (落)
- c. Relatively unproductive (2): *seng* (成), *can* (親)
- d. Very unproductive (4): *gin* (見), *zoek* (著), *zyu* (住), *sat* (實)

Now consider (11)b, whether inner aspectual elements are bound morphemes. It is not an easy question, since even though some elements have a free counterpart, it is not clear whether that is the same morpheme or a different one. There are however several clear cases. *Jyun* (完) clearly can be a free morpheme with exactly the same “finish” meaning, as in *tou hei jyun-zo* (套戲完咗) ‘the movie has ended’. In contrast, *dou2* (到), which probably comes from tone changed *dou3* (到) ‘arrive’, no longer can stand alone. *Seng* (成) comes from *sing* (成) in *singgong* 成功 ‘success’, but *sing* (成) itself is also bound. It is also clear that *zoek* (著), *can* (親), *hei* (起), *lok* (落), *zyu* (住), *ding* (定) and *gwo* (過) lack free forms. Cases fall in the middle are *gin* (見), *hou* (好), *dim* (掂), and *sat* (實), which seem to retain some but not all the lexical meaning of their free forms.

(13) Boundness of inner aspectual elements

- a. Free (1): *jyun* (完)
- b. Maybe free or bound (4): *gin* (見), *hou* (好), *dim* (掂), *sat* (實)
- c. Bound (9): *dou* (到), *zoek* (著), *seng* (成), *can* (親), *hei* (起), *lok* (落), *zyu* (住), *ding* (定) and *gwo* (過)

As for (11)c vacuity, we have already established that inner aspectual elements are different from RVCs that the former convey lexical meaning but the latter convey aspectual meaning. It is however true that some inner aspectual elements may retain a certain extent of the original lexical meaning, such as *seng* (成) from *singgong* 成功 ‘success’. *Jyun* (完), unfortunately, is simply untestable: it has aspectual meaning even in its free form.

(14) Vacuity of inner aspectual elements

- a. Aspectual, with little or no lexical meaning (9): *dou* (到), *zoek* (著), *jyun* (完), *can* (親), *hei* (起), *lok* (落), *zyu* (住), *ding* (定), and *gwo* (過)
- b. Retaining some lexical meaning: (5): *gin* (見), *seng* (成), *hou* (好), *dim* (掂), *sat* (實)

Criterion (11)d exclusivity is not considered as a valid one for reasons discussed above. Moreover, all the inner aspectual elements, as their defining property, can be followed by perfective suffix *-zo* (咗). We will see however that some should be verbal suffixes.

Criterion (11)e refers to whether the verbal root and the postverbal element can be “infixated” by *-dak/m-* 得/唔 ‘able/not’ to form the so-called potential forms, which is possible

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for verbal complements but not for verbal suffixes. As shown in (15), most inner aspectual elements allow *-dak/m-* 得/唔 infixation. The few that reject it are *hei* (起), *lok* (落), *ding* (定), and *gwo* (過). It is thus quite strong evidence that these four are verbal suffixes. The other 10 elements, in contrast, are more PC-like.

(15) Potential forms (*-dak/m-* 得/唔 ‘able/not’ infixation)⁹

- a. 有時都攞唔到，好似今日咁多人，就未必攞得到 (Video caption, 2021-2-10)
Jausi dou lo-m-dou, houci gamjat gam-do jan,
sometimes also get-not-ACHV like today that-many person
zau meibit lo-dak-dou.
then uncertain get-able-ACHV
‘Sometimes (you) cannot get it. When there are as many people like today, it’s not certain that (you) can get it.’
- b. 佢瞓{得/唔}著
Keoi fan-{dak/m}-**zoek**
3SG sleep-able/not-ACHV
‘He can(not) sleep.’
- c. 睇{得/唔}見
Tai-{dak/m}-**gin**
see-able/not-ACHV
‘can(not) see’
- d. 我估今晚去得成 (Matthews and Yip 1994:221)
Ngo gu gammaan heoi-dak-seng
1SG guess tonight go-able-ACHV
‘I think we’ll manage to go tonight’
- e. 嚇唔親你添。
Haak-m-**can** nei timl.
scare-not-ACHV 2SG SFP
‘(I) couldn’t scare you.’
- f. 食{得/唔}完
Sik-{dak/m}-**jyun**
eat-able/not-COMPL
‘can(not) finish eating’
- g. 話我一定鎖唔好隻門 (Discord, 2024-1-19)
Waa ngo jatding so-m-hou zek mun
say 1SG definitely lock-not-COMPL CL door
‘Say that I definitely cannot lock the door properly.’
- h. 搞{得/唔}掂
Gaau-{dak/m}-**dim**
do-able/not-COMPL
‘can(not) settle’

⁹ The source URL links of the internet examples are provided below. (a) and (k) were accessed on 2022-6-23, and (g) was accessed on 2024-12-24.

(a): <https://www.youtube.com/watch?v=57s8tFuEXbU>

(g): <https://www.dcard.tw/f/hkmacgirl/p/254459847>

(k): <https://www.facebook.com/HKBusChannel/posts/2078493698854987/>

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- i. *畫{得/唔}起幅畫¹⁰
*Waak-{dak/m}-**hei** fuk waa
draw-able/not-COMPL CL drawing
Int.: ‘can(not) finish the drawing’
- j. *煮{得/唔}落湯底¹¹
*Zyu-{dak/m}-**lok** tongdai
cook-able/not-COMPL soup.base
Int.: ‘can(not) cook the broth in advance’
- k. 香港啲公司到呢個年代仲以為可以畀得住啲消息 (Video caption, 2021-2-10)
Hoenggong di gungsi dou ni-go nindoi
Hong Kong CL.PL company to this-CL era
zung jiwai hoji kam-dak-zyu di siusik.
still think can cover-able-CONT CL.PL news
‘How could the HK companies think that they can cover up the news nowadays?’
- l. 揸{得/唔}實
Gam-{dak/m}-**sat**
hold-able/not-CONT
‘can(not) hold still’
- m. *煮{得/唔}定餐飯¹²
*Zyu-{dak/m}-**ding** caan faan
cook-{dak/m}-in.advance CL rice
Int.: ‘can(not) cook the meal in advance’
- n. *重新寫{得/唔}過篇文
*Cungsan se-{dak/m}-**gwo** pin man
again write-able/not-again CL paper
Int.: ‘can(not) write the article again’

Regarding the last criterion (11)f, most inner aspectual elements (n=10) cannot follow an RVC as in (16), a canonical property of being a PC.¹³ *Ding* (定) and *gwo* (過) resist *-dak/m-* infixation, and, as expected, can follow RVCs, further supporting their suffixal status. *Hei* (起) and *lok* (落), however, cannot follow RVCs, although they also resist *-dak/m-* infixation. Yet, it might be due to an independent constraint that *hei* (起) and *lok* (落) only accept monosyllabic verbal roots: compare *cyulei-jyun/hou* (處理完/好) ‘finished processing’ and **cyulei-hei/lok* (*處理起/落). The puzzling cases are *dou* (到) and *jyun* (完). While they accept *-dak/m-* infixation (i.e., PC-like), they can also follow RVCs (i.e., suffix-like). Given that they are highly productive and indicate primarily aspectual meaning, it is possible that they are mid-way to verbal suffixes.¹⁴

¹⁰ Compare: directional complement *ling-{dak/m}-hei* (拎{得/唔}起) ‘can(not) pick up’.

¹¹ Compare: directional complement *sik-{dak/m}-lok* (食{得/唔}落) ‘can(not) eat/swallow’.

¹² Compare: RVC *co-{dak/m}-ding* (坐{得/唔}定) ‘can(not) sit firmly’.

¹³ The source URL links of the internet examples are provided below, accessed on 2022-6-23.

(a):<https://www.singtaousa.com/3894174>

(f):<https://shorturl.at/aAS79>

(m):<https://filmot.com/sidebyside/HeWgxZeM1GY/zh-HK/zh-HK/Chinese+%28Hong+Kong%29/Chinese+%28Hong+Kong%29/%E3%80%90%E8%A9%A6%E7%94%A8%E8%A9%95%E6%B8%AC%E3%80%91%E5%86%87%E4%BA%BA%E8%A7%80%E5%A1%98%E5%A4%E6%B8%ACSIGMA+24mm+f3.5+DG+DN/%E8%86%A0%E6%94%9D%E7%8F%BE%E5%A0%B4>

¹⁴ It is even possible that *dou* (到) and *jyun* (完) are verbal suffixes given that another verbal suffix, *-saai* (晒),

(16) Following RVCs

- a. 後尾我哋同康華終於打開到話題。 (News, 2022-1-22)
Haumei ngodei tung Hong Waa zungjyu **daa-hoi-dou** waatai.
afterwards 1PL with Akina Hong eventually hit-open-ACHV topic
'We eventually opened up the topic with Akina Hong.'
- b. *瞓寐著
***Fan-lam-zoek**
sleep-deep-ACHV
Int.: 'slept deeply'
- c. *睇清見
***Tai-cing-gin**
see-clear-ACHV
Int.: 'saw clearly'
- d. *book 爆成呢間餐廳
***Buk-baa-u-seng** ni-gaan caanteng
book-full-ACHV this-CL restaurant
Int.: 'successfully made this restaurant fully booked'
- e. *打傷親呢個人
***Daa-soeng-can** ni-go jan
hit-hurt-ACHV this-CL person
Int.: 'beat and injured this guy'
- f. 蒸熟完就可以攪碎 (Food guide, 2014-7-18)
Zing-suk-jyun zau hoji gaau-seoi
steam-cooked-COMPL then may stir-shattered
'after finished steaming it, you can break them into pieces'
- g. ??省靚好個招牌
??saang-leng-hou go ziupaa
wipe-pretty-COMPL CL sign
Int.: 'made the fame better'
- h. *搞清楚掂
***gaau-cingco-dim**
make-clear-COMPL
Int.: 'made it clear'
- i. *寫靚起幅字
***se-leng-hei** fuk zi
write-pretty-COMPL CL word
Int.: 'finished the calligraphy artwork'
- j. *煲滾落個湯底
***Bou-gwan-lok** go tongdai
boil-boiled-COMPL CL soup.base
Int.: 'boiled the broth long time ago'
- k. *揸平住/實張紙
***gam-ping-{zyu/sat}** zoeng zi
hold-flattened-CONT CL paper
Int.: 'held and flattened the paper'

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1. 拆走定個書架 (Tang 2015:78)

Caak-zau-ding go syugaa
remove-away-in.advance CL book.shelf
'removed the book shelf in advance'

m. 真係可以再放大過呢啲位置

(Video, 2020-12-27)

Zanhai hoji zoi **fong-daai-gwo** ni di waizi.
really may again zoom-big-again this CL.PL position
'We really can re-enlarge these positions.'

Table 2 summarizes the results of the five tests on the 14 inner aspectual elements. I take *-dak/m-* infixation to be a hallmark property for phase complements, and ability to follow RVCs for verbal suffixes. There are in total 4 verbal suffixes and 8 PCs, and 2 in-between cases: *dou* (到) and *jyun* (完). The results of these two tests are generally consistent with their productivity (lower for PCs; higher for suffixes). For boundness and vacuity, there seems to be an one-way correlation: suffixes are all bound and all lack lexical meaning, whereas bound elements without lexical meaning are not necessarily suffixes (e.g., *can* 親).

Inner aspect	Productivity	Bound?	Vacuity?	<i>-dak/m-</i> infixing?	Following RVCs?	Status
<i>ding6</i> (定)	High	YES	YES	*	OK	Suffix
<i>gwo3</i> (過)	High	YES	YES	*	OK	Suffix
<i>hei2</i> (起)	Mid-high	YES	YES	*	*	Suffix
<i>lok6</i> (落)	Mid-high	YES	YES	*	*	Suffix
<i>dou2</i> (到)	High	YES	YES	OK	OK	In-between
<i>jyun4</i> (完)	High	NO	YES	OK	OK	In-between
<i>hou2</i> (好)	Mid-high	Maybe	NO	OK	*	PCs
<i>dim6</i> (掂)	Mid-high	Maybe	NO	OK	*	PCs
<i>seng4</i> (成)	Mid-low	YES	NO	OK	*	PCs
<i>can1</i> (親)	Mid-low	YES	YES	OK	*	PCs
<i>zyu6</i> (住)	Low	YES	YES	OK	*	PCs
<i>zoek6</i> (著)	Low	YES	YES	OK	*	PCs
<i>sat6</i> (實)	Low	Maybe	NO	OK	*	PCs
<i>gin3</i> (見)	Low	Maybe	NO	OK	*	PCs

Table 2: The grammatical status of inner aspectual elements

4. A formal analysis

This section offers a formal syntactic analysis of inner aspectual element. First, I assume the split-aspect approach where there are two aspectual projections: one above *vP*, another below *vP* (Gu 1995, Tsai 2008, Huang, Li, and Li 2009, Sybesma 2017, Lu, Lipták, and Sybesma 2019, Yip 2020a, Tang 2022, Lee and Pan 2024; for other languages, see MacDonald 2008, Travis 2010).

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(17) The split-aspect approach

[TP ... [AspP-outer **Asp**_{outer} [vP ... [AspP-inner **Asp**_{inner} [VP ...

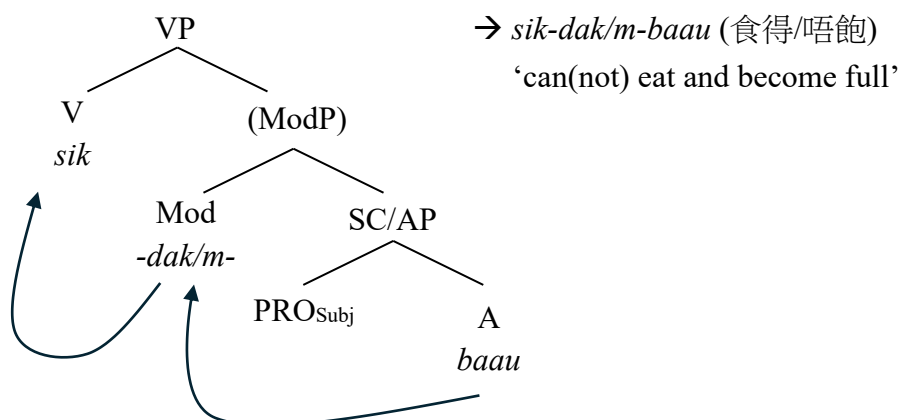
Second, I assume the non-lexicalist approach where suffixes host a functional head and project phrase, rather than being base-generated with the verb on V (Tang 1998, 2003, 2006, Tsai 2001, 2008, T. T.-M. Lee 2024b; against the lexicalist approach in Gu 1993, 1995, Huang, Li, and Li 2009). In other words, suffixes are not part of the verb in the lexicon. The suffixation is achieved by verb movement in the syntax. For example, the verbal root *sik* (食) ‘eat’ is base-generated at V and moves up to perfective *-zo* (咗) at **Asp**_{outer} to form the verb *sik-zo* (食咗) ‘ate’, as illustrated in (18).

(18) Suffixation by syntactic verb movement

- a. [TP ... [AspP-outer **-zo** [vP ... v [AspP-inner **Asp**_{inner} [VP *sik* ‘eat’ ... (baseline)
b. [TP ... [AspP-outer *sik-v-Asp*_{inner}**-zo** [vP ... [AspP-inner [VP ... (syntax)
c. *sik-∅-∅-zo* = *sik-zo* (PF)

Third, I propose that RVCs do not form a compound with the verbal root either. They are small clauses (SCs) taken by the verb, and undergoes head movement to the verb (in a manner like V-to-V incorporation). The RVC assigns theta-role to a PRO at the specifier, which is co-referential with the subject. To derive the *-dak/m-* (得/唔) infixation, I follow Cheng and Sybesma (2004) in assuming a VP-internal Modal Phrase (ModP) hosted by *-dak/m-* (得/唔). Head movement goes up in a “snowball” fashion, as in (19). I further assume the infixal status of *-dak/m-* so they always get inserted between RVCs and the verbal root.

(19) RVCs as small clauses

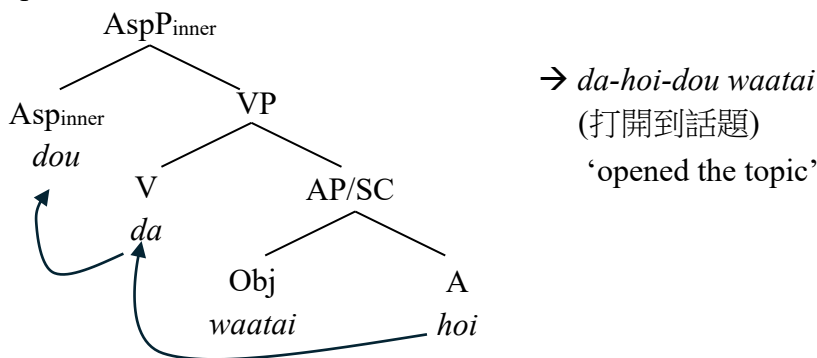


To capture the verb doubling contrasts, I propose that only lexical heads may be doubled. Functional projections, on the other hand, cannot be doubled (see T. T.-M. Lee 2024a:41 for the failure of modals to undergo topic doubling). This explains why verbs and RVCs (i.e., lexical projections such as A or V) can be doubled, whereas aspectual elements (i.e., **Asp**_{inner}/**Asp**_{outer}) cannot.

Now, we can discuss how inner aspectual elements are represented in the syntax. As for the suffixal elements, I assume that they are base-generated at the **Asp**_{inner} head and attract head movement. In the case with RVCs, V moves up to **Asp**_{inner} and carry along the RVC. Hence, the inner aspectual suffixes follow the RVCs. When the object is present, it is base-generated at the specifier of the SC, and receives theta-roles from both the RVC and the V. The derivation is illustrated in (20).

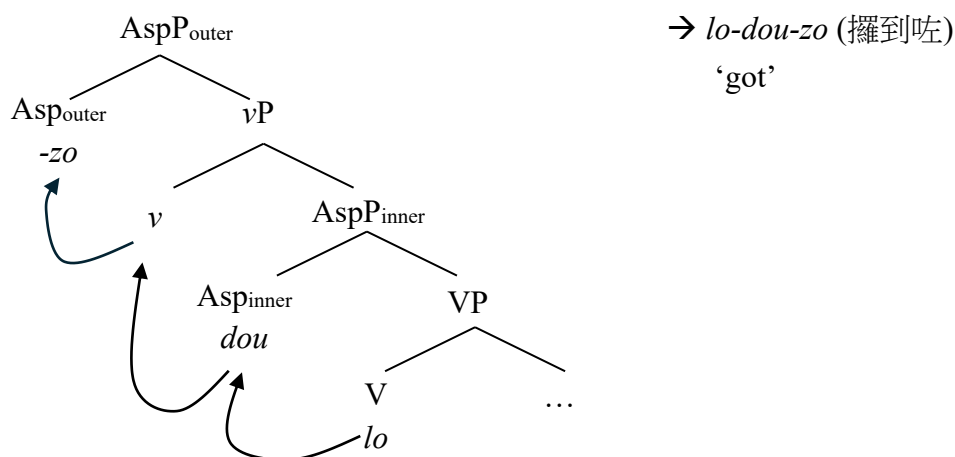
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(20) Inner aspectual suffixes follows RVCs



On the other hand, inner aspectual suffixes are within vP . When outer aspectual suffixes are also present, the V moves along with Asp_{inner} all the way to Asp_{outer} above vP . Hence, inner aspect necessarily follows outer aspect, capturing *-zo* suffixation.

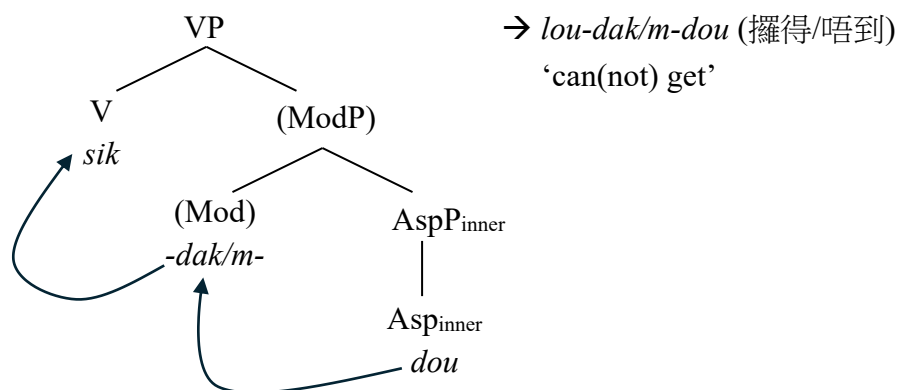
(21) Inner aspectual suffixes precedes outer aspectual suffixes



The question remains is how phase complements like *zoek* (著) are derived. One straightforward option is to treat them like RVCs and base-generate the SC under VP . There are however two problems. First, SCs are lexical projections like VP and AP , but PCs should host aspectual projections for their meaning, and to capture the resistance to be doubling (i.e., functional projections cannot be doubled). Second, SCs assign theta-roles to the object and/or a PRO co-referential to the subject. PCs nevertheless does not assign any theta-roles.

To resolve this dilemma, I propose that there is a third aspectual projection below VP (see Tsai 2008 for similar ideas). This is illustrated in (22):

(22) Phase complement project a VP-internal



In this way, the PC moves up to Mod and then V to derive the *-dak/m-* (得/唔) infixation. On the other hand, the PC projects an aspectual head and cannot be doubled. Note that for inner aspectual elements that have the intermediate status between suffixes and PCs, I suggest that they may be projected either above VPs (following RVCs) or below VPs (with *-dak/m-* 得/唔 infixation).¹⁵

5. Conclusion

To conclude, this paper has provided a detailed and systematic description of the properties of the inner aspectual elements in Cantonese. In total 14 such elements have been identified, supported by diagnostics including *-zo* (得) suffixation and failure to undergo verb doubling. With comprehensive investigation of the properties of these elements, including productivity, boundness, vacuity, potential forms (*-dak/m-* 得/唔 infixation) and ability to follow, it is concluded that 4 of them are verbal suffixes: *ding* (定), *gwo* (過), *hei* (起), and *lok* (落); 8 of them are phase complements: *zoek* (著), *gin* (見), *seng* (成), *can* (親), *hou* (好), *dim* (掂), *zyu* (住), and *sat* (實); and 2 of them have an intermediate status: *dou* (到) and *jyun* (完).

I also offered a formal syntactic analysis. Adopting a non-lexicalist approach, neither verbal complements nor suffixes are base-generated with the verbal root in the lexicon, but they enter syntax as different units. Suffixes host functional projections above VP, where two aspectual projections are available: Asp_{outer} and Asp_{inner} . The verb undergoes syntactic head movement to the aspectual suffix to form a suffixed verb. On the other hand, resultative verbal complements form small clauses that are taken by the verb, below VP. Phase complements also project $AspP_{inner}$ below VP. Both RVCs and PCs move to V, and potentially take up the infixal *-dak/m-* 得/唔 on an optional VP-internal modal projection. In this way, we obtain the formal basis for three distinctions: (i) the verbal complement-suffix distinction: whether they are base-generated VP-internally or externally; (ii) the outer-inner aspect distinction: whether they are base-generated vP -internally or externally; (iii) the resultative-aspectual distinction: whether they head lexical projections or functional (aspectual) projections.

¹⁵ The current structure predicts that PCs may be stackable with inner aspectual suffixes. I leave this issue to further research.

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